The Covariant Superstring on K3

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Introduction

Covariant superstrings

Compactification

Dimensional reduction of the pure spinor

Conclusion

Introduction

- We want to use superstrings as a calculus for quantum field theories coupled to quantum gravity.
- The simplest definition of this calculus is as a free ten-dimensional string.
- To use this tool for lower-dimensional physics, we need to reduce from 10D.
- First method: Holographic (Brenno's talk here last week)
- Second method: Compactification
- The compactification problem of relating the pure spinor in 10D to the RNS string in lower dimensions has resisted solution since 2002.
- We will solve this problem for compactification of the pure spinor string to six dimensions on a K3 surface (*i.e.* the four-dimensional Calabi-Yau manifold).

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Unfortunately the flat space limit of this string is only free in light-cone gauge so covariant quantization is not straightforward.

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Siegel proposed the following modification:

$$S_{\mathrm{GSS}} \sim S_{\mathrm{GS}} + \int_{\Sigma} d\partial\theta,$$

where the Siegel constraint is given by

$$d = p_\theta + i(\theta\gamma^m)\partial x_m + \theta^2\partial\theta\text{-term}.$$

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So quantization gives the canonical operator product algebra

$$x(z)x(0) \sim \log|z|^2, \quad p_\theta(z)\theta(0) \sim \frac{1}{z}.$$

Dirac quantization

Then it is easy to compute the operator product

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Problem: To use the Dirac procedure, we have to separate the constraints. Is there a Poincaré-covariant way to do this?

Berkovits' formalism I (1994): Hybrid

Consider the RNS string on a factorized background

$$\begin{array}{ccc} M & \to & M \times X \\ \mathrm{Hol}(\nabla_M) & \to & \mathrm{Hol}(\nabla_M) \times \mathrm{Hol}(\nabla_X) \end{array}$$

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	"space-time"	"internal"
matter	x^m, ψ^m	$y^i, \bar{y}^{\bar{\imath}}, \psi^i, \bar{\psi}^{\bar{\imath}}$
ghosts	b, c, β, γ	
	$\downarrow$	$\downarrow$
Berkovits variables	$x^m, p_\theta, \theta, \rho, \sigma$	$y^i, \bar{y}^{\bar{\imath}}, \psi^i, \bar{\psi}^{\bar{\imath}}$

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and a factorized algebra of constraints

$$\mathrm{SCA}_{10} \to \mathrm{SCA}_{\text{space-time}} \oplus \mathrm{SCA}_{\text{internal}}.$$

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Proceed with BRST-type quantization!

Mysteries

- What is the pure spinor? Harmonic? Ghost? Twistor? Mix?
- What is the relation with the hybrid formalism?
- What is the relation (if any) with the pure spinors of generalized geometry?
- How to recover the correct spectrum in compactification with off-shell superspace?
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Strategy: Compactify on K3 which has the advantages that it

- is simple,
- has a hybrid description,
- has harmonics in its superspace,
- can be compactified further to 4 dimensions on T^2 to make K3-fibered Calabi-Yau 3-fold.

K3 orbifold

Start with a torus (*very* special holonomy)

$$T^4 \approx \frac{\mathbb{C}\times\mathbb{C}}{\mathbb{Z}\times\mathbb{Z}},$$

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with $2^4 = 16$ fixed points which can be blown up to Eguchi-Hanson spaces $\mathcal{O}_{\mathbb{P}^1}(-2) \approx T^*S^2$ (gravitational instantons).

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$$\begin{aligned} E &= \underbrace{6(-\tfrac{1}{24}) + 4(\tfrac{1}{48})}_{x+\tilde{x}} + \underbrace{8(\tfrac{1}{12}) + 8(-\tfrac{1}{24})}_{\theta,p+\tilde{\theta},\tilde{p}} + \underbrace{n(-\tfrac{1}{12}) + (11-n)(\tfrac{1}{24})}_{\lambda,w+\tilde{\lambda},\tilde{w}} \\ &= -\frac{1}{8}(n-5). \end{aligned}$$

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Spectrum matches iff $n = 5 \Rightarrow 6$ of the original 11 pure spinors are $\mathbb{Z}_2$-twisted.

$U(5)$ variables

Breaking $Spin(10) \to U(5)$, we can solve the pure spinor constraint

$$\begin{array}{ccccccc} \lambda & \to & \gamma & & \gamma u_{[ab]} & & \gamma u_{[ab} u_{cd]} \\ \mathbf{16} & = & \mathbf{1} & \oplus & \overline{\mathbf{10}} & \oplus & \mathbf{5} \end{array}$$

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Small Hilbert Space

The Berkovits differential splits

$$Q = \oint \left(\lambda^A u^a d_{Aa} + \lambda_{Aa'} d^{Aa'} + \Lambda^A d_{A2} \right) = Q_0 + Q_1 + Q_2$$

were Q_m has m factors of $\mathbf{6}$ in it.

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Claim:

$$H_Q(C^\bullet) = H^0_{Q_0}(C^\bullet).$$

In words: The physical states are described by the cohomology of a short differential on a small Hilbert space without **6**-variables.

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are both differentials. Here, $d_A^+ = u^a \left(p_{Aa} + i\theta_a^B \partial x_{AB} + \epsilon_{ABCD} \theta_a^B \theta^{Cb} \partial \theta_b^D \right)$ and $\psi_{aa'} = \gamma\, u_a\, \theta_{1a'}$.

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By Dolbeault's theorem, the cohomology of the double complex becomes the relative cohomology

$$H_{Q_0} \approx H_\delta(H_q).$$

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Meanwhile, H_δ is the ring of vertex operators of the form $\omega_{aa'bb'\ldots}\psi^{aa'}\psi^{bb'}\ldots$ which are annihilated by $\delta = \psi^{aa'}\partial_{aa'}$ and not δ-exact.

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Meanwhile, H_δ is the ring of vertex operators of the form $\omega_{aa'bb'\ldots}\psi^{aa'}\psi^{bb'}\ldots$ which are annihilated by $\delta = \psi^{aa'}\partial_{aa'}$ and not δ-exact.

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Hence, $H_\delta(H_q)$ is the cohomology algebra of K3 vertex operators with projective superfield wave functions.

Projective superspace

The space-time part of the hybrid formalism on K3 lives in projective superspace

$$\mathbb{R}^{6|8} \times \mathbb{C}P^1 \quad \approx \quad \frac{\mathbb{R}^{6|8} \rtimes (Spin(6) \times SU_R(2))}{Spin(6) \times U(1)_R}.$$

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Vertex operators have superspace wave functions $\Phi(x, \theta, u)$, which can be expanded in harmonics

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The Siegel constraints act, by the operator product, as superspace derivatives

$$d_{\alpha i}(z)\Phi(x, \theta, u) = \frac{1}{z} D_{\alpha i}\Phi(x, \theta, u).$$

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acts as D^+, implying that string wave functions are projective superfields.

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To ensure only ρ/σ-dependence in the combination u, vertex operators must have no poles with the constraint $J = \partial(\rho + i\sigma)$ (which generates shifts $(\rho, \sigma) \mapsto (\rho + a, \sigma + ia)$).

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This terminates the expansion and puts the wave function on-shell. The constraints $\{J, G^\pm, T\}$ generate the desired $N = 2$ worldsheet superconformal algebra of the *hybrid* (a.k.a. RNS) superstring on K3.

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Fin

Thank you!

The work reported here was done in collaboration with Osvaldo Chandía and Brenno Carlini Vallilo in the following papers:

O. Chandia, Wm D. Linch III and B. C. Vallilo, "Compactification of the Heterotic Pure Spinor Superstring I," JHEP **0910**, 060 (2009) [arXiv:0907.2247 [hep-th]].

O. Chandia, Wm D. Linch III, B. C. Vallilo, "Compactification of the Heterotic Pure Spinor Superstring II," JHEP **1110**, 098 (2011). [arXiv:1108.3555 [hep-th]].

O. Chandia, Wm D. Linch III, B. C. Vallilo, "The Covariant Superstring on K3," [arXiv:1109.3200 [hep-th]].